ADIDAS US DATA ANALYSIS

BUSINESS PERFORMANCE CASE STUDY

Commercial Performance & Channel Efficiency

A comprehensive evaluation of Adidas US sales data (2020–2021) examining category performance, consumer demographics, channel profitability, and regional distribution dynamics.

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5.1 Strategic Recommendations

1. Business Task Statement

1. Statement of the Business Task

The objective of this project is to analyze Adidas US sales data to identify key performance drivers across product categories (Footwear vs. Apparel), consumer segments (Men's vs. Women's), and sales channels (Online, In-store, Outlet).

The insights generated will help optimize inventory allocation, maximize operating margins, and improve channel strategies for a sportswear retail distributor.

2. Guiding Business Questions

- **Category Breakdown:** Do shoes or apparel generate higher total sales and operating profits?
- **Demographic Segment Performance:** How do sales compare between Men's and Women's product lines?
- **Channel Profitability:** Which sales method (Online, In-store, or Outlet) yields the highest operating margin?
- **Geographic & Retailer Demand:** Which regions and retail partners drive the highest sales volume and profitability?

3. Key Stakeholders

- **Executive Leadership / Head of Retail Strategy:** Primary stakeholder focused on high-level profitability and expansion strategies.
- **Inventory & Supply Chain Manager:** Secondary stakeholder focused on product demand and stock allocation.
- **E-Commerce & Marketing Director:** Secondary stakeholder focused on channel performance and segment marketing.

4. Key Metrics (KPIs)

Total Sales Revenue, Operating Profit, Operating Margin, Units Sold, Average Selling Price / Unit, Sales Growth by Channel, Sales Growth by region, Category Revenue Split, Retailer Contribution Rate.

2. Source

1. Data Source Details

Dataset Name: Adidas US Sales Dataset
Source / Origin: Kaggle (Public dataset curated by Heemali Chaudhari)
Format: Single CSV file (Adidas US Sales Datasets.csv)
Timeframe: Covers two full years of transaction records (2020 – 2021).
Storage Location: Raw data is securely stored locally in a dedicated project directory and backed up in cloud storage prior to processing.

2. Dataset Structure & Schema

The dataset contains 9,648 rows (transaction records) across 13 attributes:

Column Name	Data Type	Description
Retailer	Text / String	Name of the retail partner (e.g., Foot Locker, Walmart, Amazon)
Retailer ID	Numeric / ID	Unique identifier for the retailer
Invoice Date	Date	Date of the transaction (YYYY-MM-DD)
Region	Text / String	Geographic region in the US (e.g., Northeast, South, West)
State	Text / String	US State where the sale occurred
City	Text / String	US City where the sale occurred
Product	Text / String	Product category (e.g., Men's Street Footwear, Women's Apparel)
Price per Unit	Currency / Float	Unit price of the product sold ($)
Units Sold	Integer	Total quantity of units sold in the transaction
Total Sales	Currency / Float	Total transaction value (Price per Unit × Units Sold)
Operating Profit	Currency / Float	Profit generated after operating expenses ($)
Operating Margin	Percentage / Float	Operating profit margin (%)
Sales Method	Text / String	Channel method (In-store, Outlet, Online)

3. Data Integrity & Credibility Assessment (ROCCC Analysis)

- **Reliable (High):** The dataset provides consistent transactional structures with standard monetary, temporal, and categorical variables across 9,000+ entries.
- **Original (Medium):** Second-party dataset aggregated on Kaggle based on US retail historical performance.
- **Comprehensive (High):** Contains full regional coverage of the US, multiple distribution channels, detailed category descriptions, and financial profit metrics.
- **Current (Medium):** Covers historical data through 2021, providing a solid baseline for multi-year trend analysis and retail benchmarking.
- **Cited (High):** Sourced from a public repository with clear origin tags and wide adoption within the data community.

4. Data Licensing & Privacy (Ethics)

Privacy / Licensing: Publicly available open dataset. It contains no Personally Identifiable Information (PII) regarding end-customers or sensitive individual enterprise records.

3. Documentation of Data Cleaning

1. Tool Selection & Justification

- **Microsoft Excel:** Selected for initial data inspection, sanity checks, format verification, and basic error detection.

2. Data Cleaning & Verification Steps

To ensure data integrity, reliability, and readiness for analysis, the following cleaning steps were executed:

Initial Inspection & Format Validation:

- Opened the raw dataset in Excel to verify structural consistency.
- Checked column headers to ensure clean naming conventions (no leading/trailing spaces).
- Validated that Invoice Date was properly formatted as a date data type (YYYY-MM-DD).

Handling Missing & Null Values:

- Inspected primary key columns (Retailer ID) and transaction metrics for missing entries.
- *Result:* The dataset was verified to have zero critical null values across essential transaction rows.

Data Type Enforcement & String Standardization:

- Verified quantitative metrics were configured as numeric/currency data types.
- Screened categorical columns using Excel filters and SQL constraints (`LOWER() / TRIM()`) to eliminate duplicate categories caused by typography or casing inconsistencies.

Anomaly & Outlier Check: Scanned for logical anomalies (e.g., negative units sold, zero pricing errors). No corrupted records were found.

3. Verification of Clean Data Output

The processed dataset was saved as a clean `.csv` file (`adidas_usa_sales.csv`) inside the `02_processed_data` directory, fully optimized for database ingestion and visualization tool loading.

4. Executive Summary of Analysis

$899.90M	$332.14M	36.91%
TOTAL REVENUE	OPERATING PROFIT	OPERATING MARGIN

1. Executive Summary & Core Financial KPIs

During the evaluated period, Adidas US generated $899.90M in total revenue with an operating profit of $332.14M, maintaining a healthy overall operating margin of 36.91% across 2.48M units sold at an Average Selling Price (ASP) of $363.03.

Key Takeaway: The business exhibits strong profitability with solid margins exceeding 36%, driven by high sales volume in core footwear categories and key strategic retail partnerships.

2. Product Category Performance & Gender Demographics

- **Footwear Dominance:** Men's Street Footwear is the single largest revenue driver, generating $208.83M (23.21% share) with an operating margin of 39.65%.
- **Gender Nuances:**
  - *Men:* Highly skewed toward footwear (Men's Street Footwear + Men's Athletic Footwear combined represent 40.29% of total revenue).
  - *Women:* Show higher engagement in apparel. Women's Apparel is the #2 overall revenue contributor ($179.04M / 19.90% share) and outperforms Women's Street Footwear ($128.00M / 14.22% share).
- **Overall Category Mix:** Footwear accounts for the majority of sales volume and revenue compared to apparel.

3. Sales Channel Performance & Margin Efficiency

- **In-Store:** Leads in total revenue ($356.64M / 39.63% share) due to foot traffic and physical presence, maintaining a 35.78% operating margin.
- **Outlet:** Represents $295.59M (32.85% share) with a 36.53% margin.
- **Online:** Accounts for $247.67M (27.52% share) but yields the highest operating margin at 38.99%.

Insight: Online sales generate higher efficiency and profitability per unit, likely due to lower fixed operational overhead (no physical store staffing, lease, or utility costs).

4. Retailer Contribution & Distribution Dynamics

- **Top Performers (Footwear Specialists):** West Gear ($242.96M / 27.00% share) and Foot Locker ($220.09M / 24.46% share) collectively drive over 51% of total Adidas US revenue. Their athletic focus and extensive brick-and-mortar footprint maximize brand reach.
- **Underperformers (Big-Box / General Merchants):** Amazon ($77.70M / 8.63% share) and Walmart ($74.56M / 8.29% share) sit at the bottom of revenue contribution. Footwear is not their primary core product niche.
- **Margin Dynamics:** Retailers with a stronger digital/direct push (e.g., Sports Direct at 40.74% margin) yield better operating returns.

5. Geographic Performance & Regional Distribution

- **Top Revenue Region:** The West region leads overall sales, generating $269.94M (30.0% share) and 686.98K units sold, though operating at a 33.20% margin.
- **Highest Margin Region:** The South region, while generating $144.66M in revenue, achieved the highest operational efficiency with a 42.26% operating margin.
- **Regional Breakdown:** West ($269.94M / 33.20%), Northeast ($186.32M / 36.51%), Southeast ($163.17M / 37.11%), South ($144.66M / 42.26%), Midwest ($135.80M / 38.89%).

💡 Strategic Recommendations

- **Capitalize on Online Margin:** Expand digital marketing and direct-to-consumer (DTC) e-commerce capabilities to capture more of the high-margin (38.99%) digital market.
- **Double Down on Core Retail Partners:** Deepen co-marketing and inventory allocation with West Gear and Foot Locker, while reassessing the strategy for Amazon/Walmart to improve inventory turnover.
- **Cross-Sell Apparel to Men & Footwear to Women:** Leverage the strength of Men's Footwear to bundle Men's Apparel, and create promotional bundles combining Women's Apparel with Women's Footwear.

Dashboard Overview

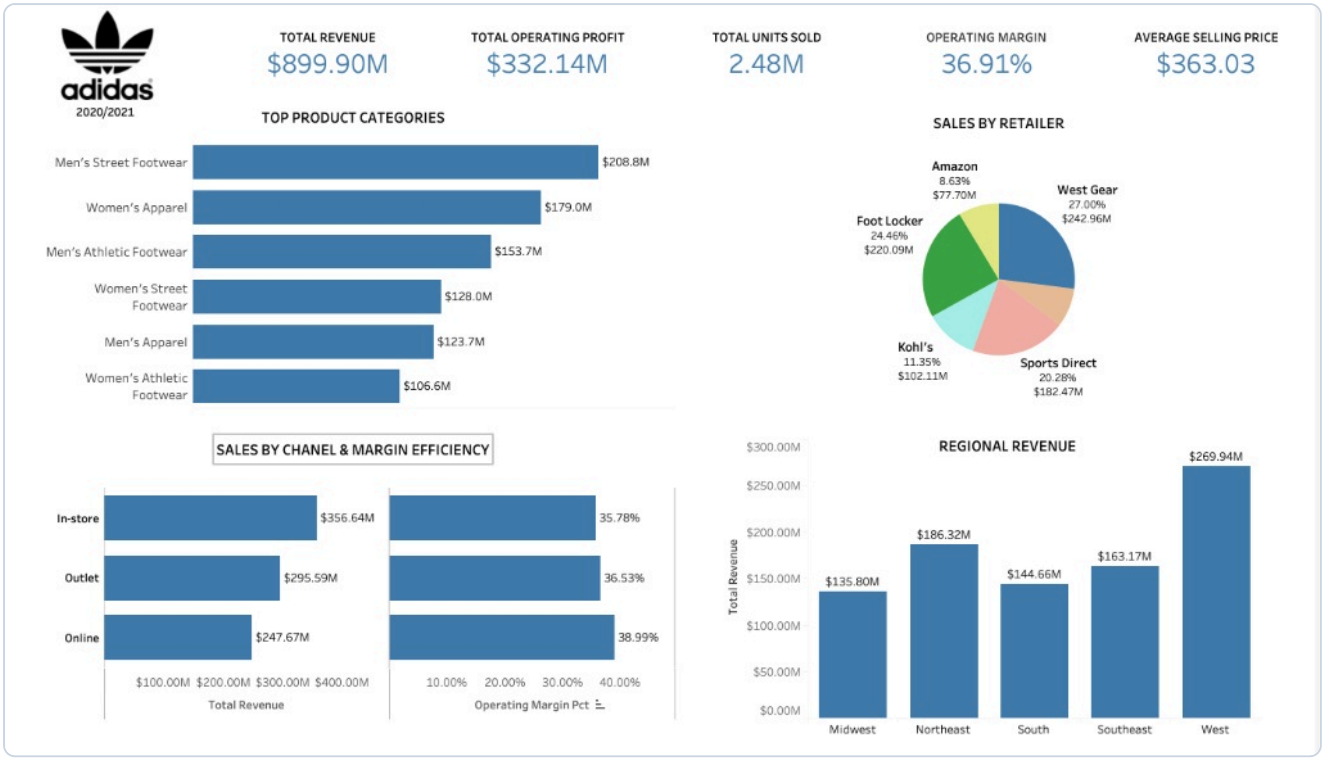


Figure 1: Executive Dashboard mapping Category Breakdown, Retailer Share, Channel Margin Efficiency, and Regional Sales Revenue.